Convex Functions: Domain

Function:

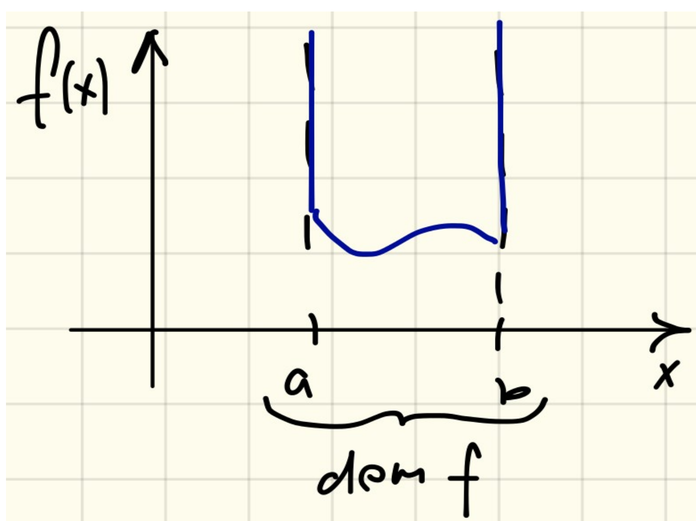
$$A \subseteq \mathbb{R}^n \qquad f : A \to B \qquad \xrightarrow{\text{Def}} \qquad \operatorname{dom} f := A$$

$$B \subseteq \mathbb{R}$$

Extended-value function:

$-\infty$ $\qquad\qquad\qquad\qquad\qquad\qquad$ $-\infty <$

$$f : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\} \qquad \xrightarrow{\text{Def}} \qquad \operatorname{dom} f := \{x \in \mathbb{R}^n : f(x) < \infty\}$$



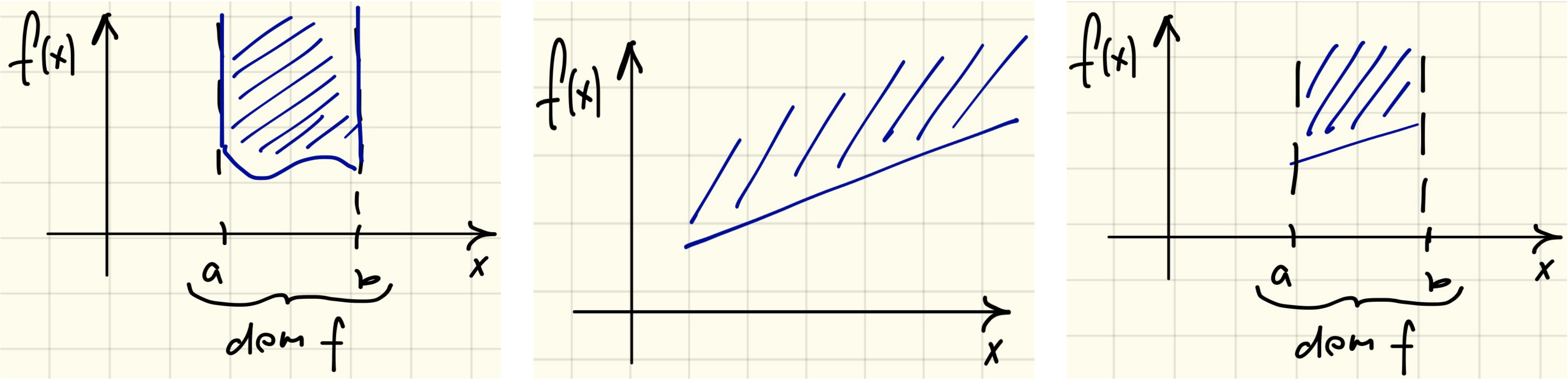
Epigraph:

$$f : A \to B$$

$$f : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$$

$\xrightarrow{\text{Def}}$

$$\mathrm{epi} f := \{(x,t) : x \in \mathrm{dom} f,\ t \geq f(x)\}$$



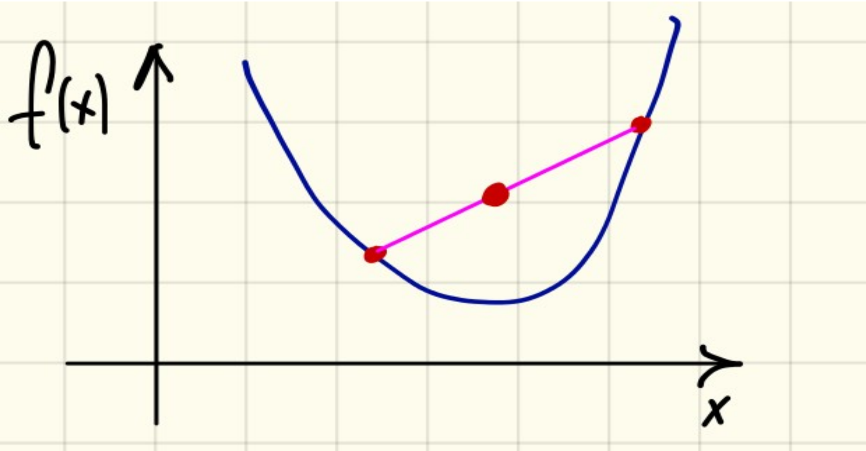
Convex Functions: Definition

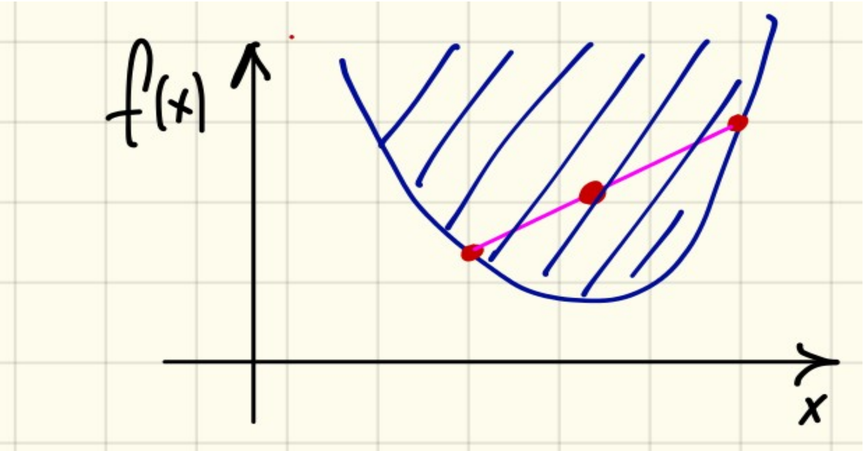
Convex function:

$$\mathrm{epi}\, f - \text{ convex set} \xrightarrow{\text{Def}} f - \text{ convex function}$$

Convex function:

$$f\left(\sum_{i=1}^{n} p_i x^i\right) \leq \sum_{i=1}^{n} p_i f(x^i) \;\forall n,\; p \in \Delta^n$$





Convex Functions: Definition

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Concave function:

$$(-f) - \text{ convex function} \xrightarrow{\text{Def}} f - \text{ concave function}$$

$$(f : \mathbb{R}^n \to \mathbb{R} \cup \{-\infty\})$$

Concave function:

$(-f)-$ convex function $\xrightarrow{\text{Def}}$ $f-$ concave function

$$(f : \mathbb{R}^n \to \mathbb{R} \cup \{-\infty\})$$

Epigraph of a concave function is:

1) concave
2) convex
3) neither concave nor convex in general
4) none is correct

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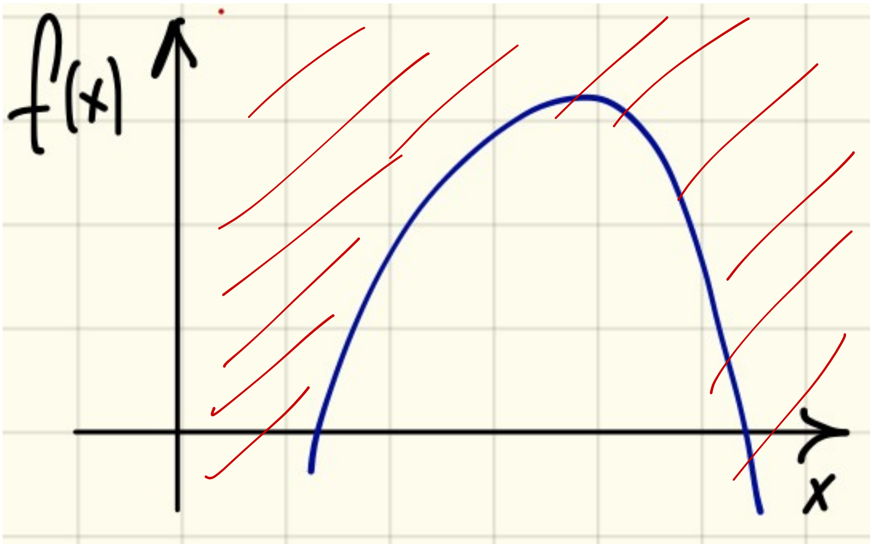

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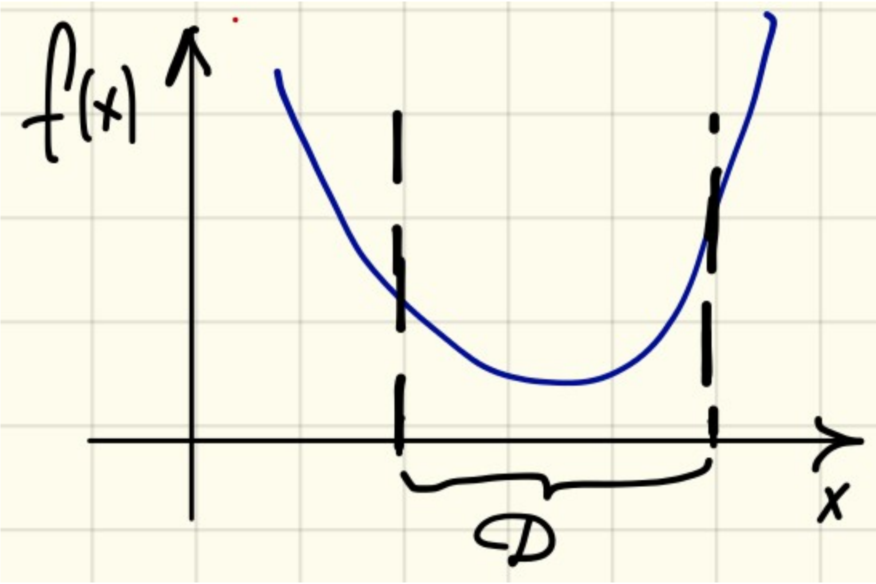
Linear function:

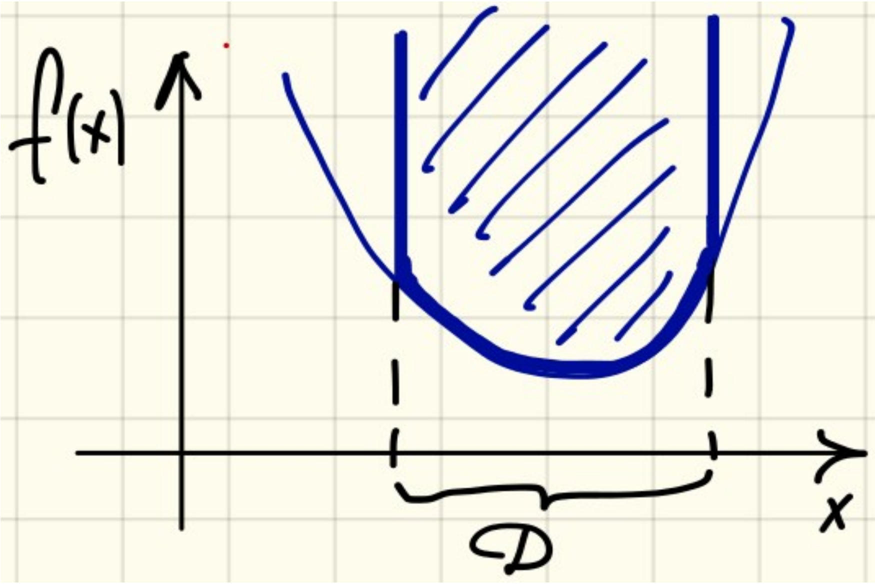
$$f - \text{ convex and concave} \quad \xrightarrow{\text{Def}} \quad f - \text{ linear}$$

Convex function on a convex set of constraints = convex:

$$f : D \to \mathbb{R}$$
$$f - \text{ convex}$$
$$D - \text{ convex}$$

$$\Rightarrow \quad \tilde{f} = \begin{cases} f(x), & x \in D \\ \infty, & x \notin D \end{cases} - \text{convex}$$



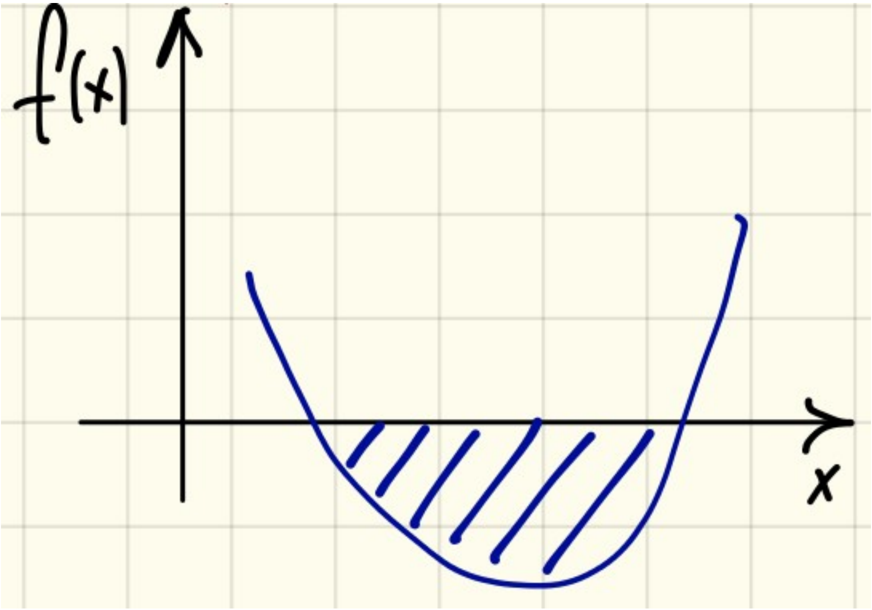


Convex Functions: Examples

Proposition:

$$f : \mathbb{R}^m \to \mathbb{R} \cup \{\infty\} - \text{ convex} \quad \longrightarrow \quad \begin{array}{l} D := \{x : f(x) \leq 0\} \\ D - \text{ convex} \end{array}$$

$$D = \mathbb{R}^m \cap \operatorname{epi} f$$



Convex Functions: Examples

Intersection of hyperplanes is convex: $\{x : Ax = b\}$– convex set

$$p \in \Delta^n \qquad A\sum_{i=1}^{n} p_i x^i = \sum_{i=1}^{n} p_i A x^i = \underbrace{\sum_{i=1}^{n} p_i}_{=1 \text{ by def.}} b = b$$

Here could be your illustration!

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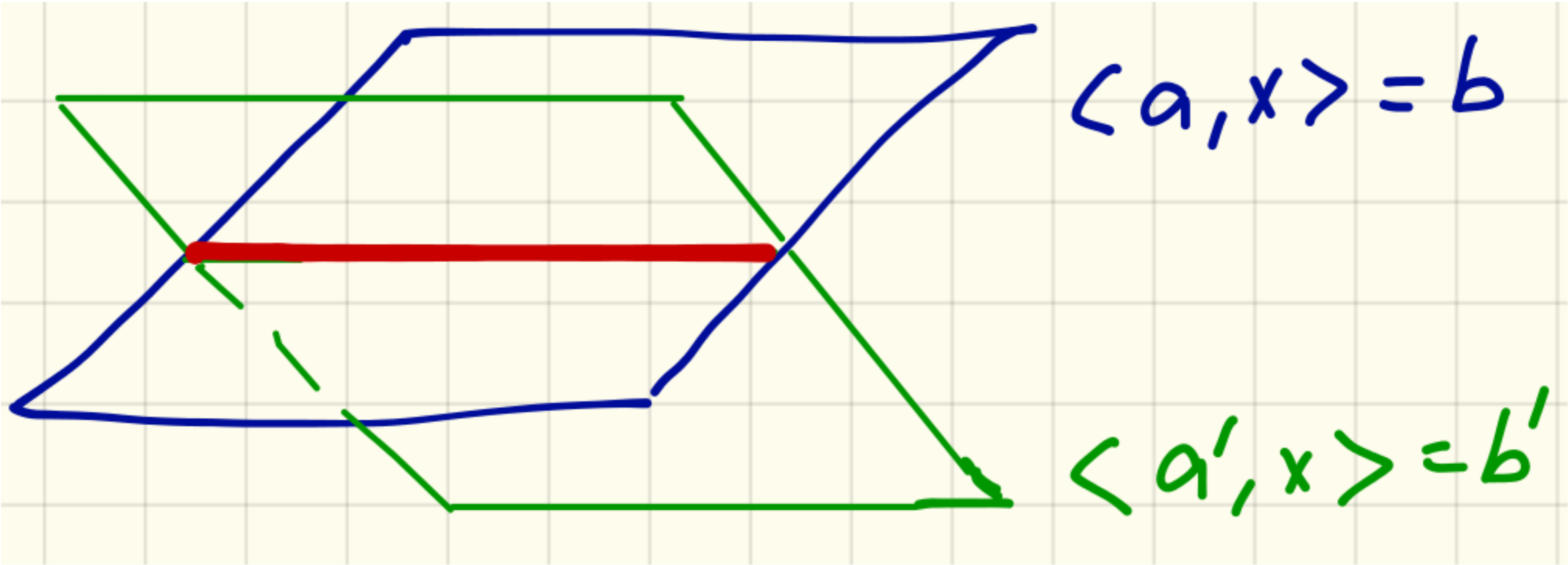


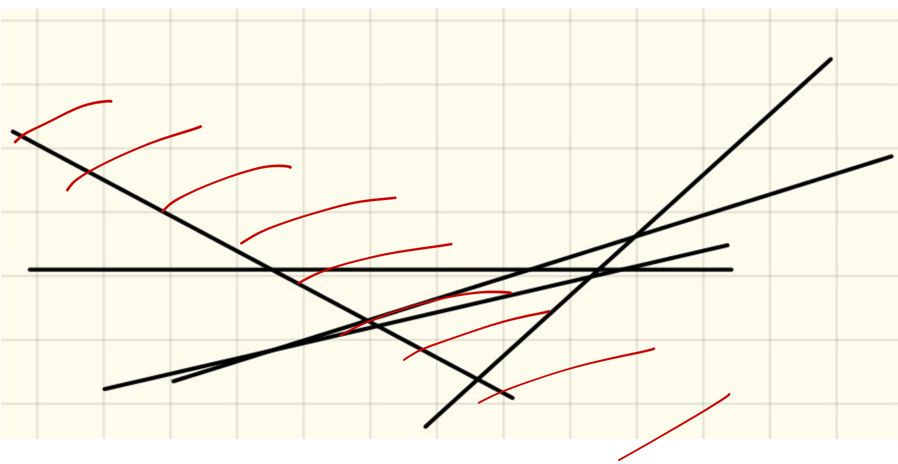
Convex Functions: Examples

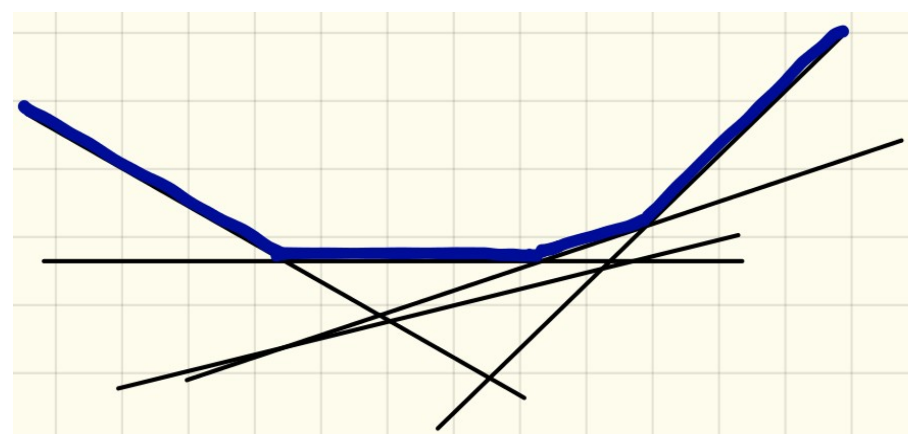
Supremum of linear functions is convex:

$Y \subseteq \mathbb{R}^n$

$f : \mathbb{R}^n \times Y \to \mathbb{R}$ $\longrightarrow$ $g(x) = \sup_{y \in Y} f(x, y)$– convex

$f(\cdot, y)$– linear, convex





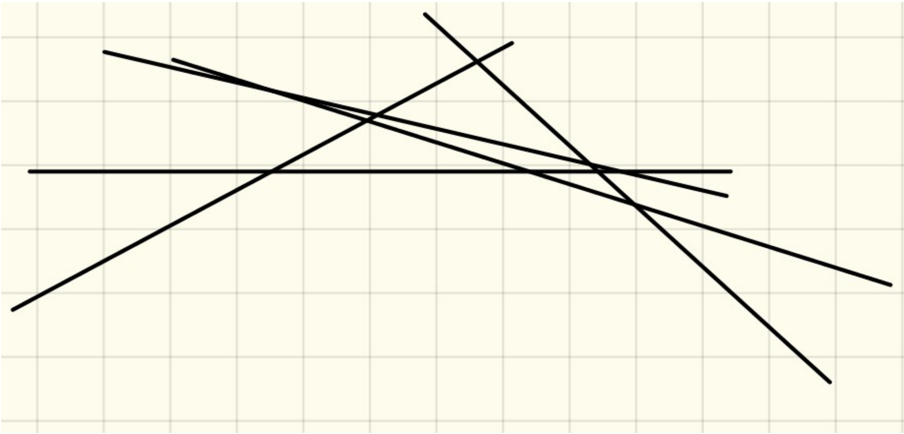
Convex Functions: Examples

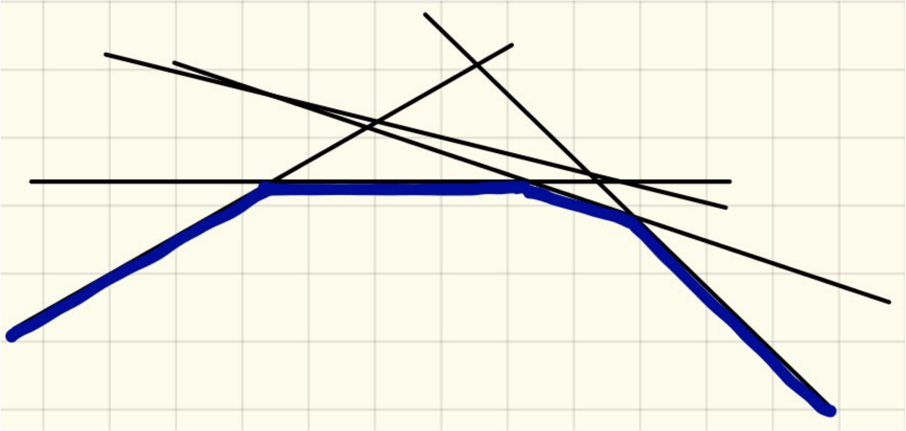
Infinum of linear functions is concave:

$$Y \subseteq \mathbb{R}^n$$

$$f : \mathbb{R}^n \times Y \to \mathbb{R} \quad \longrightarrow \quad g(x) = \inf_{y \in Y} f(x, y) - \text{concave}$$

$$f(\cdot, y) - \text{linear}$$, concave





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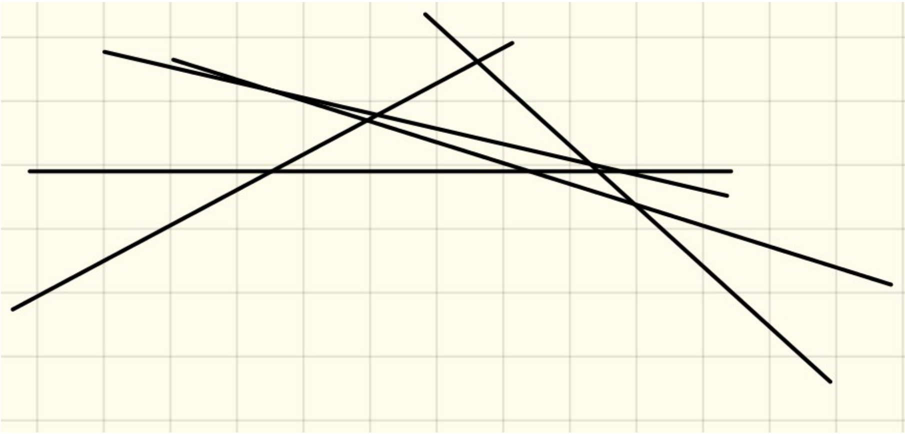

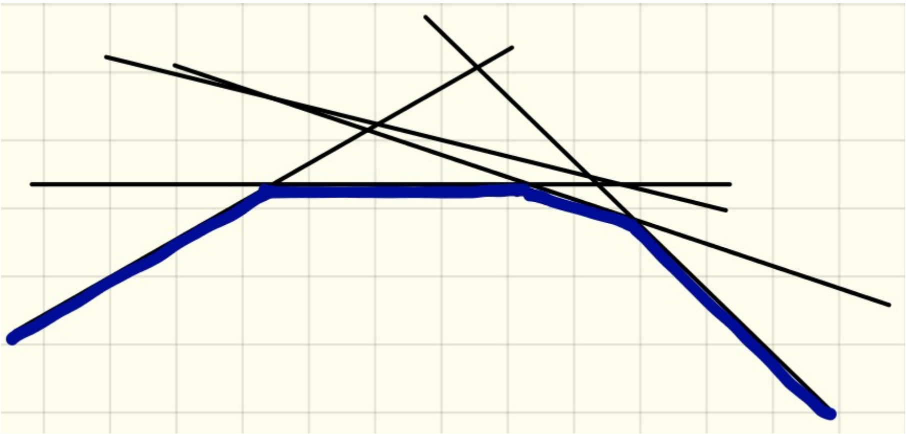

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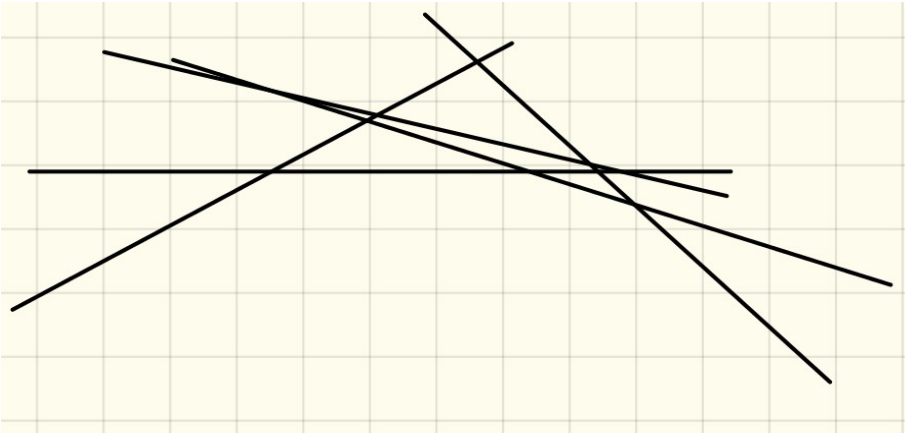

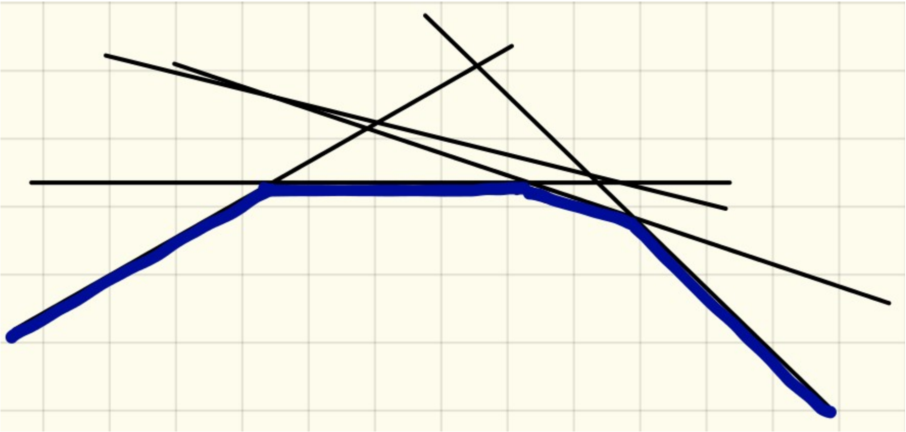

Why convex functions are important for optimization?

→ local optimum = global optimum

→ gradient of the function - local property shows in the direction of the optimum

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